// GRACE W.
TEO

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PROFESSIONAL SUMMARY

Senior UI/UX designer and human factors researcher with more than 15 years of experience partnering with product managers, engineers, and business stakeholders to translate complex user needs and business objectives into actionable insights that shape product strategy, design, feature prioritization, and organizational decision-making in technology-heavy domains. Strong track record of collaborating within cross-functional Agile teams and communicating solution concepts to technical and non-technical audiences.

EXPERIENCE

Quantum Improvements Consulting, Orlando, FL

Lead Scientist (Senior Human Factors & UX Researcher): Apr 2020 – Feb 2026

- Led research strategy and execution across multiple concurrent, multi-year product initiatives with budgets ranging from $300K to over $2M. Partnered with designers, engineers, and stakeholders to reduce uncertainty, prioritize investments, and translate user insights into product direction and evaluation frameworks.
- Influenced product strategy by identifying high-value features based on user needs, prioritizing multimodal information presented to users, and partnering with engineers to define data quality requirements and validation checkpoints that improved product reliability.
- Collaborated on the research and UI design of a multimodal team training platform for tablet devices that enabled instructors to evaluate teams coordinating and executing tasks across dynamic operational environments. Identified user information needs, prioritized multimodal data streams (e.g., GPS, audio, IMU), developed behavioral metrics, and co-designed data visualizations that enabled instructors to rapidly interpret team performance.
- Directed discovery research that informed features for a collaborative scenario-planning platform supporting distributed decision-making, identifying user needs related to trust, shared understanding, and team coordination.
- Developed and executed evaluation metrics and strategies for AR/VR training solutions, leading pilot studies that demonstrated a 15% improvement in procedural knowledge and 100% positive user engagement compared with traditional training methods, helping secure stakeholder support for broader adoption.
- Accelerated qualitative data processing with AI-assisted workflows (Whisper AI) and identified behavioral measures and patterns most critical for team performance with applied machine learning techniques.
- Mentored 8–10 junior researchers on study design, analysis, and research best practices, strengthening research quality across multiple programs.

Institute for Simulation and Training, University of Central Florida, Orlando, FL

Research Associate (Human Factors): Aug 2013 – Mar 2020

- Led research for the adaption strategy of a human-agent teaming system by modeling cognitive workload and developing of an individualized workload models and decision thresholds identifying high-demand states using physiological measures (e.g., eye-tracking, EEG, ECG, fNIR, TCD).
- Translated findings of Human–Robot Interaction (HRI) research on users' mental models and natural communication patterns into design implications for more intuitive voice- and command-based interfaces.
- Contributed to US. Nuclear Regulatory Commission (NRC)-funded research on workload and error risk in simulation interfaces, identifying tradeoffs between field-of-view fidelity and response time, and input modality effects on selection accuracy to inform interface design decisions.
- Developed reusable research frameworks and standardized methodologies for study execution and analysis, improving consistency and scalability of research across projects.
- Communicated research findings and design recommendations to technical and non-technical stakeholders and mentored junior researchers in study design, data collection, and analysis.

University of Central Florida, Orlando, FL

Human Factors Researcher: Aug 2008 – Jul 2013

- Collaborated on research to inform the design of adaptive driver assistance systems by analyzing driver behavior and cognitive load under varying conditions, generating requirements for system adaptation and feedback mechanisms.
- Conducted research on operator vigilance in high-stakes environments, identifying cognitive and behavioral factors that informed system design considerations for attention and workload management.
- Designed and executed studies on vigilance and performance in high-workload environments, identifying implications for attention management, alert design, and human–system interaction under sustained demand.

DSO National Laboratories, Singapore

Research Scientist (Human Factors): Oct 2005 – Aug 2008

- Led research on decision-making and performance under stress in cyber operations environments, identifying cognitive and workflow constraints that impacted operator effectiveness in high-tempo scenarios.
- Analyzed operational workflows and cognitive demands in cybersecurity tasks, translating findings into recommendations for improving task design and decision support under time-critical conditions.

Civil Service College / Public Service Division, Singapore

Senior Psychologist / Psychologist (Industrial-Organizational Psychology): Apr 1996 – Oct 2005

- Evaluated and administered psychometric assessments to support personnel selection and workforce decision-making, ensuring validity and reliability of selection tools used in hiring processes.
- Designed and analyzed organizational surveys supporting leadership development and workforce planning initiatives, translating workforce data into insights for talent and organizational decisions.

EDUCATION

Ph.D., Applied Experimental & Human Factors Psychology: University of Central Florida, 2015
M.A., Applied Experimental & Human Factors Psychology: University of Central Florida, 2013
B.S. (Honors), Psychology: National University of Singapore, 1996

CERTIFICATES & PROFESSIONAL DEVELOPMENT

Data Mining with SAS Certificate: University of Central Florida, 2013
Design for Usability Certificate: University of Central Florida, 2012
IBM AI Product Management Certificate (in progress): Coursera
Google UX Design Certificate (in progress): Coursera

SKILLS

UX Research

Research lifecycle • Mixed-methods research • Data & measurement frameworks • Quantitative & qualitative synthesis • Generative & evaluative studies • Psychometrics • Power analysis • Benchmarking • Statistical analysis • Task analysis • A/B testing • Behavioral analytics • Physiological measures (e.g., eye-tracking) • Experiments • Interviews • Surveys • Diary studies • Navigation testing • Journey mapping • Personas • Think-aloud protocol • Cognitive walk-throughs • Heuristic evaluations • User flows & interaction design • Usability testing • Requirements & data architecture support

Leadership & Collaboration

Research leadership • Product strategy • Systems thinking • Human-AI collaboration • Agile product development • Cross-functional collaborations • Stakeholder management • Executive presentations • Workshop facilitation

Tools & Technologies

Python • R • SQL • SPSS • Power BI • Qualtrics • Figma • UserTesting • JIRA • MS Office Suite • AI-based tools (e.g., Whisper AI for speech-to-text transcriptions; Ollama + Mistral + Python for an offline AI agent)